

Managing Duplicate Data

Duplicate data is a common issue in real-world datasets. **Duplicates** can arise for several reasons: manual data entry errors; merging datasets from multiple sources; or system glitches that cause repeated records. The presence of duplicate rows can distort your analysis by inflating counts; skewing statistical summaries; and leading to incorrect conclusions. Removing duplicates is a crucial step to ensure the accuracy and reliability of your data-driven insights.

```
import pandas as pd

# Sample DataFrame with duplicate rows
data = {
    "name": ["Alice", "Bob", "Alice", "David", "Bob"],
    "age": [25, 30, 25, 22, 30],
    "city": ["New York", "Paris", "New York", "London", "Paris"]
}
df = pd.DataFrame(data)

# Identify duplicate rows
duplicates = df.duplicated()
print("Duplicated rows:")
print(duplicates)

# Remove duplicate rows
df_no_duplicates = df.drop_duplicates()
print("\nDataFrame after removing duplicates:")
print(df_no_duplicates)
```